

PAPER_714: UQFF Knowledge Base 18: THz Q-Scope Signal Analysis (Set 50: Signals 41-50)

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Class: UQFFKnowledgeBaseKB18 **CP4 Entry:** #298 **Keywords:** THz, q-scope, signals 41-50, oscilloscope, 200ns, 500mV, flow state, UQFF KB18 **Session:** 175 | **Version:** v5.32
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Abstract

UQFF Knowledge Base 18 (KB18) presents the detailed analysis of THz q-scope Set 50 (Signals 41-50), captured at 16:48:23–16:50:20 UTC-4 on the Star-Magic q-scope apparatus. All 10 signals exhibit 1.246 THz fundamental with the characteristic Ch1/Ch2 dual-channel signature used to classify ACE/DCE flow state.

1. Instrument Configuration (Set 50)

Parameter	Value
Time/Div	200 ns
Volt/Div	500 mV
Channels	Ch1 (sensor A), Ch2 (sensor B)
Duration	117 s (16:48:23–16:50:20)
Signals	41–50 (10 total)

2. Signal Amplitude Data

Signal	Ch1 (V)	Ch2 (V)	Flow State
41	0.60	0.35	Normal
42	0.65	0.40	Normal
43	0.60	0.35	Normal
44	0.55	0.30	Chaotic
45	0.50	0.35	Inverted
46	0.60	0.40	Inverted
47	0.55	0.35	Chaotic
48	0.50	0.30	Chaotic
49	0.50	0.35	Inverted
50	0.50	0.35	Inverted

3. UQFF Signal Energy

$$E_{signal}(i) = \frac{V_{eff,i}^2}{Z} \Delta t, \quad Z = 50 \Omega, \quad V_{eff} = V_{ch1}/\sqrt{2}$$

Bundle sum (weighted by flow state):

$$B_{set50} = \sum_{i:fs \neq 0} E_i \cdot fs(i)$$

4. U_m Contribution

$$U_{m,i} = \mu_J \omega_{THz} V_{ch1,i} (1 - e^{-\kappa(t+t_i)})$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Q-Scope Laboratory Records Set 50, Star-Magic 2026
- UQFF Framework v5.32, Star-Magic Session 175

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